GraphCompose

v2.1.0

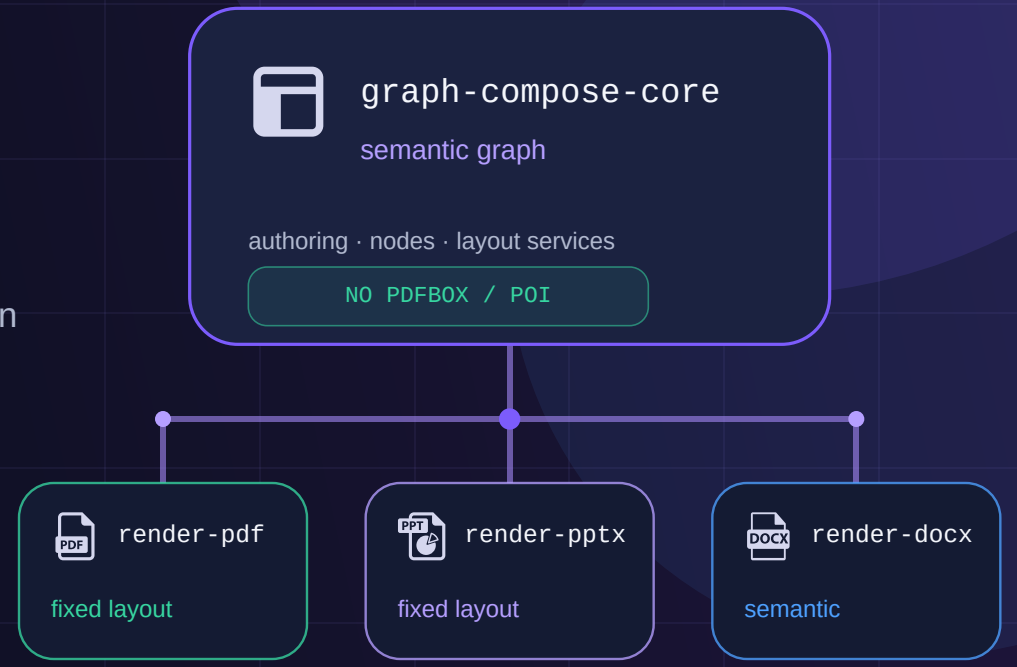
GRAPHCOMPOSE / MODULE-FIRST

Compose once.
Render through modules.

A lean Java document core with a semantic model, deterministic layout geometry and opt-in output backends.

```
io.github.demchaav:graph-compose-core
+ choose a renderer module
```

THE MODULE GRAPH

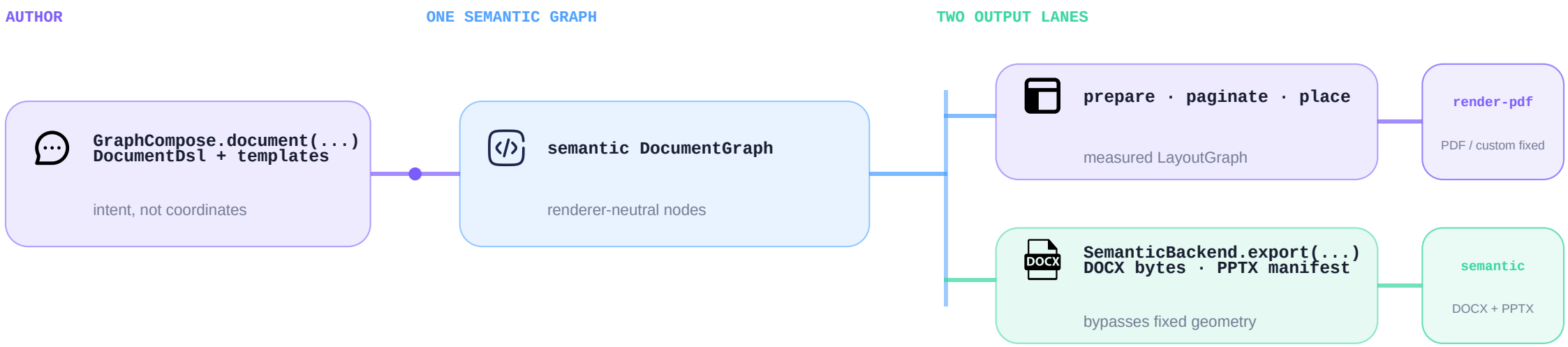


- `graph-compose-core` — lean engine
- `graph-compose` — drop-in PDF
- `ServiceLoader SPI` — bring a renderer
- `layoutSnapshot()` — geometry regression

01 / ARCHITECTURE

Semantic at the core. Format at the edge.

The canonical DSL produces one semantic DocumentGraph. Fixed-layout output resolves geometry; semantic exporters consume the graph directly.



Deterministic geometry is testable before bytes are rendered through layoutSnapshot().

LEAN CORE
No PDFBox, POI, zxing or template families in graph-compose-core.

FIXED BACKEND SPI
FixedLayoutBackendProvider and FontMetricsProvider are discovered by ServiceLoader.

SEMANTIC EXPORT
DOCX is a working POI exporter; PPTX carries a slide-safe skeleton beside its fixed backend.

The renderer boundary is explicit; Android or another backend is an extension opportunity, not a shipped 2.0 claim.

02 / PACKAGING

Choose only what you need.

Eight engine-train artifacts move in lockstep. Fonts and emoji keep independent version lines; heavy render and template dependencies are opt-in.

DROP-IN PDF DEFAULT

graph-compose

Core + PDF renderer. Canonical callers keeping this coordinate retain PDF out of the box.

LEAN / CUSTOM LEAN

graph-compose-core

Authoring, semantic model and layout services. Add a renderer module only when needed.

BATTERIES INCLUDED BUNDLE

graph-compose-bundle

Default PDF stack plus templates, bundled fonts and emoji. Office exporters stay opt-in.

THE LOCKSTEP TRAIN

One version, smaller dependency trees, explicit format boundaries.

- core — authoring + model
- render-pdf — fixed PDF
- render-docx — semantic DOCX
- render-pptx — fixed PPTX
- templates — opt-in presets
- testing — snapshot tools
- graph-compose — PDF wrapper
- bundle — PDF + templates + assets

Accepted migration trade-off: built-in templates are no longer carried by graph-compose; add graph-compose-templates or use the bundle.

03 / REFERENCE RUN

Measured locally. Read with context.

Equivalent 1000-row report fixtures from the bundled 2026-07-26 single-machine snapshot. These values are directional, not a final 2.0 marketing benchmark.

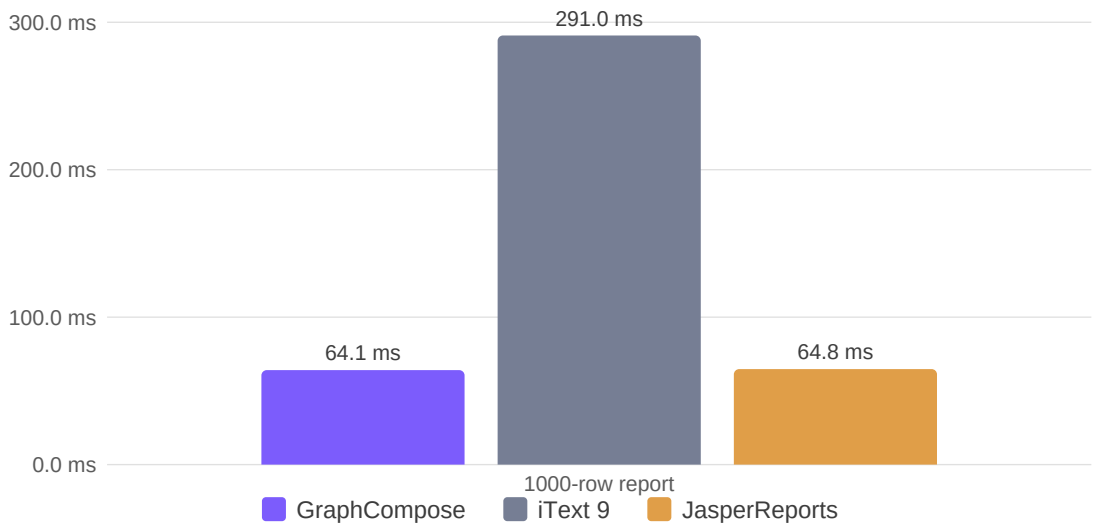
64.1 ms
GraphCompose avg render
`1000 rows`

19.2 MiB
avg allocated / render
`thread allocation`

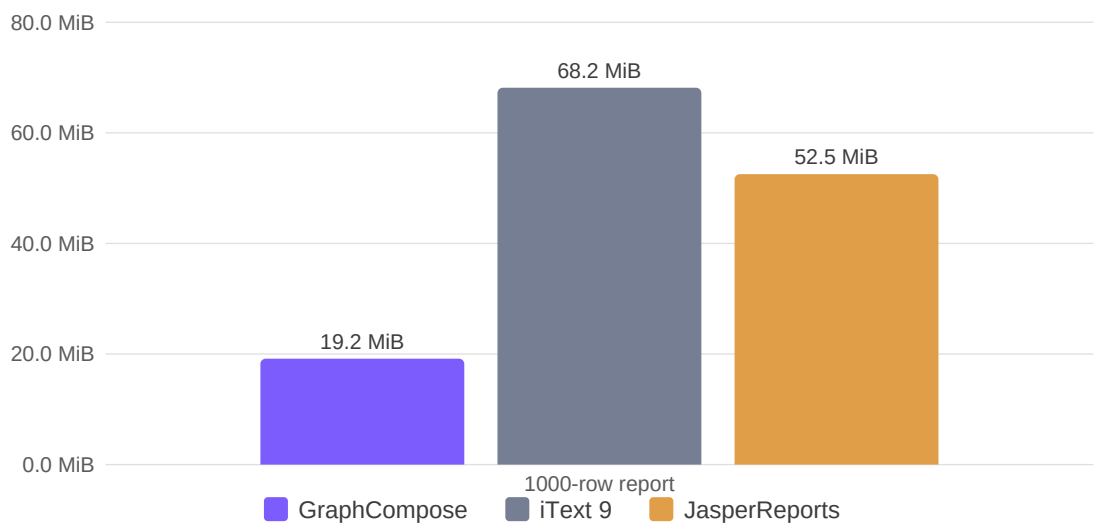
40 / 200 / 1000
report rows
`scaling fixtures`

2026-07-26
snapshot date
`provenance incomplete`

Average render latency at 1000 rows - ms



Average thread allocation per render - MiB



Scope: one developer machine, single run. The file records neither CPU, OS and JDK nor the commit measured, and its iteration counts cover only the small-invoice tier. Absolute milliseconds therefore travel badly between machines - read the ratios, not the timings.

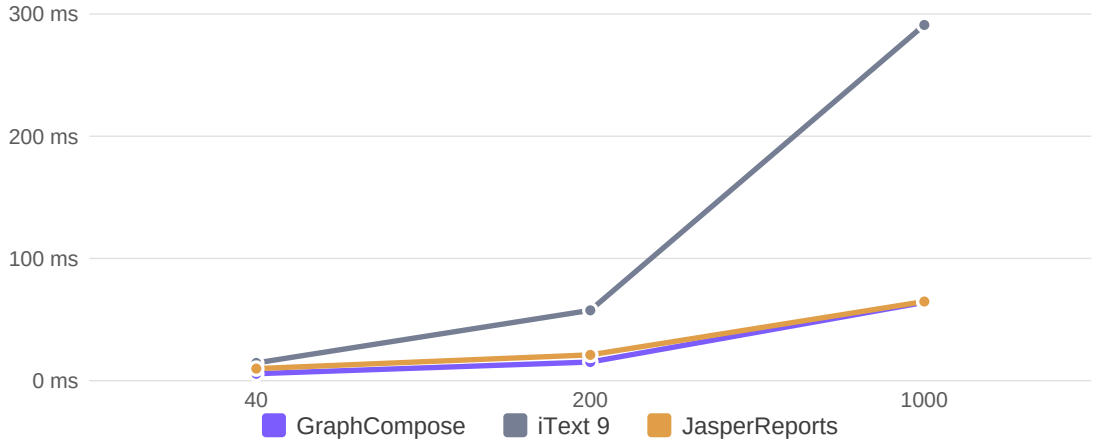
Reproduce: scripts/run-benchmarks.ps1 - Current docs: docs/operations/benchmarks.md

04 / SCALING PROFILE

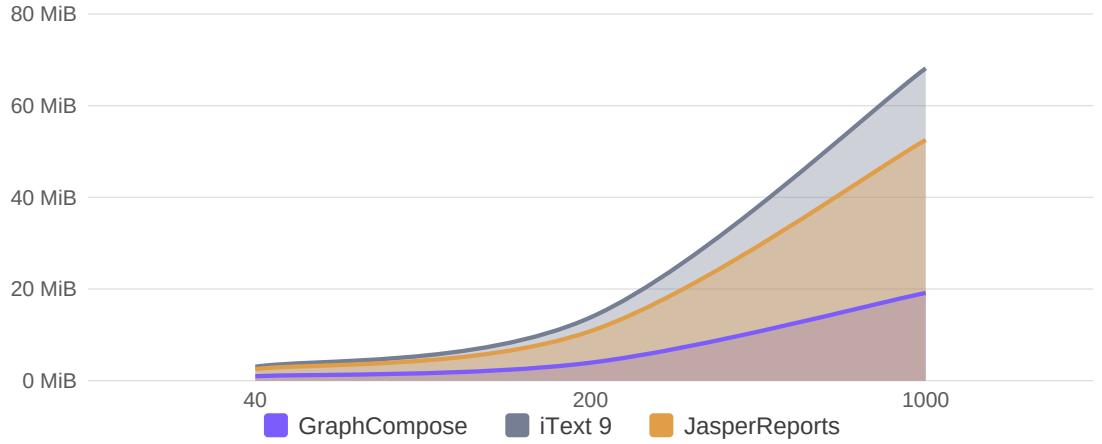
Watch the curve, not one headline number.

The same local reference run across 40, 200 and 1000 rows. Allocation values are independent per-render measurements; they are not shares of a common heap.

Average render latency vs. report size - ms



Average allocation vs. report size - MiB / render



4.54x
iText 9 took this many times longer in the 1000-row fixture

3.56x
iText 9 allocated this many times more in the same fixture

1.1%
latency gap vs JasperReports - inside the documented 5-10% local noise band

What the 2.x line can claim today: **modular packaging, renderer isolation and deterministic layout snapshots**. The figures above are one machine's reference run, refreshed with the data file.

GraphCompose renders this deck itself: gradients, canvas layers, SVG, clipping and native charts.